ODMA–URA research record

Initial ODMA-aware decoding

From the structured signal model to factor-graph BP/EP

Stage	1 of 4
Updated	30 August 2026
Status	Historical stage; algebra retained, early comparisons interpreted conservatively

This report is a chronological research record. Measured results, interpretations, and open hypotheses are labelled separately. Detailed numerical ledgers remain in `results/`.

1 Question at this stage

The project began with a receiver-side question: if the transmitted codebook has an on–off division multiple access (ODMA) structure, can a decoder use that structure more effectively than a generic global sparse-recovery method? The first implementation therefore kept the ODMA construction explicit and concentrated on message passing, activity priors, multiplicities, and fading.

The enduring contribution of this stage is the factor-graph algebra. The main empirical outcome was less favourable: the specialised variants were fragile and generic global recovery was usually stronger in the tested regimes. That result motivated the later separation of *codebook geometry* from *decoder design*.

2 Structured URA model from first principles

Let n be the number of channel uses, Q the number of ODMA placement patterns, d the local codeword length, and V the number of local codewords. A local alphabet is

$$C = [c_0, \dots, c_{V-1}] \in \mathbb{C}^{d\times V},$$

and pattern q is an embedding or linear operator $R_q \in \mathbb{C}^{n\times d}$. Message (q, v) uses

$$\phi_{q,v} = R_q c_v \in \mathbb{C}^n, \qquad \|\phi_{q,v}\|_2^2 = 1.$$

When every pair is legal, the global alphabet has $M = QV$ messages and

$$\Phi = [R_0 C \mid R_1 C \mid \cdots \mid R_{Q-1} C] \in \mathbb{C}^{n\times M}.$$

With K_a active devices, let $a_{q,v} \in \mathbb{Z}_+$ count how many devices selected message (q, v). Then

$$\sum_{q,v} a_{q,v} = K_a, \qquad y = \sum_{q=0}^{Q-1}\sum_{v=0}^{V-1} a_{q,v} R_q c_v + z = \Phi a + z,$$

where $z \sim \mathcal{CN}(0, \sigma^2 I_n)$. Device identities never appear: only the unsourced message multiset is recoverable. For known multi-antenna channel vector $h \in \mathbb{C}^{M_{\text{ant}}}$, the observation is $Y = (\Phi a)h^T + Z$; matched channel combining reduces it to the same scalar resource model with an adjusted noise level.

The payload bits label a column. They are *not* binary channel symbols: each entry of $\phi_{q,v}$ can be real or complex and therefore can carry amplitude and phase.

3 Factor graph

Flatten (q, v) into variable index $j \in \{0, \dots, M-1\}$ and write $A = \Phi$. A variable node represents the discrete count $a_j \in \{0, 1, \dots, K_a\}$; a resource factor represents

$$y_i = \sum_{j=0}^{M-1} A_{ij} a_j + z_i.$$

ODMA makes many A_{ij} exactly zero, so the graph connects a variable only to the resources occupied by its pattern. This is computationally attractive, but it also creates short loops and groups many columns into the same subspace.

3.1 Gaussian resource update

Suppose the message from variable k to resource i has mean $m_{k\to i}$ and variance $v_{k\to i}$. Excluding candidate variable j, approximate the interference by

$$\mu_{i\to j} = \sum_{k\neq j} A_{ik} m_{k\to i}, \qquad \nu_{i\to j} = \sigma^2 + \sum_{k\neq j} |A_{ik}|^2 v_{k\to i}.$$

Conditional on $a_j = t$, resource i contributes the log-likelihood

$$\ell_{i\to j}(t) = -\frac{|y_i - \mu_{i\to j} - A_{ij}t|^2}{\nu_{i\to j}} - log\nu_{i\to j} + \text{const}.$$

This is an expectation-propagation or Gaussian-interference approximation: the candidate count stays discrete while the sum of all other users is compressed to two moments.

3.2 Discrete variable update

Let $p_j(t)$ be the activity/count prior. The cavity distribution sent from variable j to resource i is

$$q_{j\to i}(t) \propto p_j(t) \exp\left(\sum_{r\in\mathcal{N}(j)\setminus i} \ell_{r\to j}(t) \right).$$

The outgoing moments are

$$m_{j\to i} = \sum_{t=0}^{K_a} t q_{j\to i}(t), \qquad v_{j\to i} = \sum_{t=0}^{K_a} t^2 q_{j\to i}(t) - m_{j\to i}^2.$$

At termination, include every neighbouring resource and choose either the posterior mean or MAP count. Damping $m^+ = \alpha m^{\text{old}} + (1-\alpha)m^{\text{new}}$ was essential because the graph is loopy.

3.3 Priors and the global load constraint

For low collision, a Bernoulli approximation uses $\Pr(a_j > 0) \approx K_a/M$. A Poisson prior with rate K_a/M allows multiplicity. The exact independent-message experiment instead induces a multinomial count vector with $\mathbf{1}^T a = K_a$; treating its coordinates independently loses this negative dependence. Several variants tried to restore the total mass by estimating activity, projecting counts to sum to K_a, or adding load feedback.

4 What was varied

The many historical variants reduce to four modelling choices:

Choice	Principle	Main risk exposed
Basic BP/EP	Gaussian interference plus a discrete count prior	Loopy feedback and overconfident messages
Load-aware updates	Enforce or estimate $\mathbf{1}^T a = K_a$	Oracle load can hide activity-estimation difficulty
Joint fading/activity	Infer channel and support together	Bilinear scale/phase ambiguity and zero-mean collapse
Pilot/pre-equalised variants	Supply extra channel structure	Changes the communication model and receiver knowledge

These are not four independent final decoders. They are successive attempts to repair a shared approximation. Retaining every code variant in the main narrative obscures that common structure, so the superseded implementation notes have been removed while this algebra and the result transition are preserved.

5 Observed behaviour and diagnosis

Evidence status: The early records are qualitative and do not provide the later seed-controlled job contract.

The specialised receiver could work well in easy configurations, but gains were not stable as load, fading uncertainty, or graph coupling increased. Global decoders—especially NNOMP with known K_a—performed better in the decisive comparisons. The likely causes were not one isolated coding bug:

1. Gaussian interference messages are least accurate when a few structured neighbours dominate a resource.
2. Reused placement masks produce highly correlated local subproblems and short graph loops.
3. Independent coordinate priors approximate, rather than impose, the global count constraint.
4. Joint fading and activity is bilinear; without pilots or a reference convention it is not identifiable in the same way as the known-channel model.
5. A more complicated structured decoder can still lose to a global method if the dominant difficulty is choosing the correct support among all M messages.

6 Decision and transition

The next research question became: *is ODMA losing because its active columns are intrinsically poor once known, or because its support is hard for the practical decoder to find?* That required a common global model, matched dense and ODMA codebooks, and an oracle-support control. Report 2 develops that decomposition.